2–42 Use current division in Figure P2–42 to obtain an expression for v_L in terms of R, R_L, and i_S.

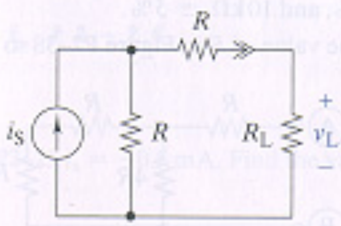

FIGURE P2–42

2–43 Find i_x in Figure P2–43.

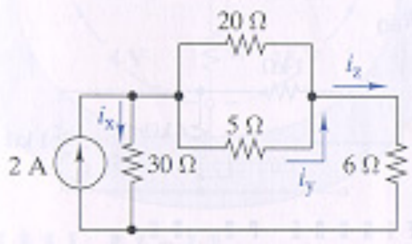

FIGURE P2–43

2–45 Find the range of values of v_O in Figure P2–45.

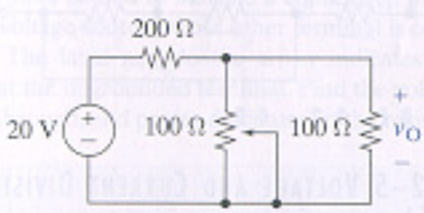

FIGURE P2–45

2–49 D Select a value of R_x in Figure P2–49 so that $v_L = 3$ V.

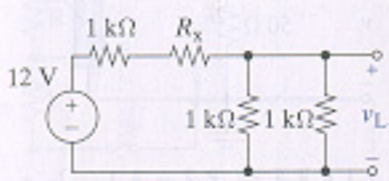

FIGURE P2–49

2–51 Use circuit reduction to find v_x and i_x in Figure P2–51.

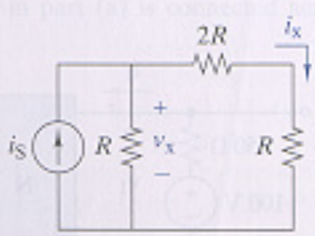

FIGURE P2–51

2–52 Use circuit reduction to find i_x in Figure P2–52.

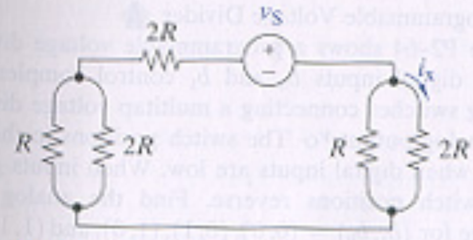

FIGURE P2–52

2–56 Use source transformation to find i_x in Figure P2–56.

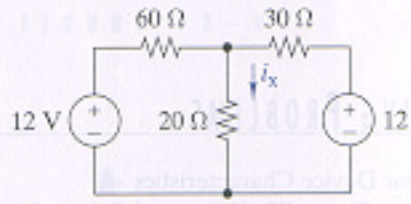

FIGURE P2–56

3–37 Find the Thévenin equivalent circuit seen by R_L in Figure P3–37. Find the voltage across the load when $R_L = 5\ \Omega$, $10\ \Omega$, and $50\ \Omega$.

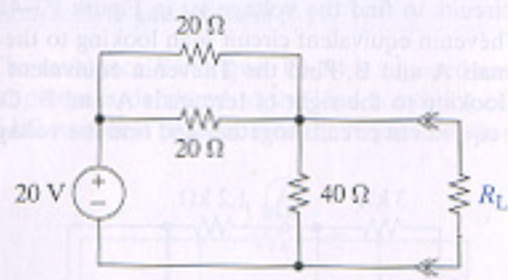

FIGURE P3–37

3–38 Find the Norton equivalent seen by R_L in Figure P3–38. Find the current through the load when $R_L = 6\ \Omega$, $12\ \Omega$, and $60\ \Omega$.

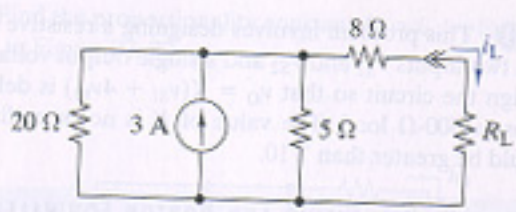

FIGURE P3–38

3–41 The purpose of this problem is to use Thévenin equivalent circuits to find the voltage v_X in Figure P3–41. Find the Thévenin equivalent circuit seen looking to the left of terminals A and B. Find the Thévenin equivalent circuit seen looking to the right of terminals A and B. Connect these equivalent circuits together and find the voltage v_X.

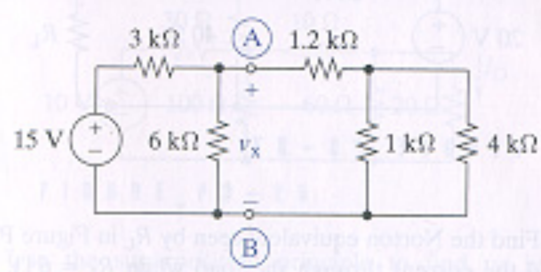

FIGURE P3–41